PRACTICAL 6

AIM: Combining and appending datasets using merge() or bind_rows() in R.

>COMMANDS USED

```
library(dplyr) # Load the library for bind_rows
and data manipulation

# -----------------------------------------
# 1. SETUP: Load and Prepare the Dataset
# -----------------------------------------
# Load the main dataset
data_main <-
read.csv("AI_Impact_on_Jobs_2030.csv")

# We will work with a sample of unique job titles
for cleaner demonstration
# Taking the first 5 unique jobs:
```

```
data_unique_jobs <- data_main %>%
  distinct(Job_Title, .keep_all = TRUE) %>%
  slice(1:5)

print("--- Base Data Sample (First 5 Unique Jobs)
---")
print(data_unique_jobs)
cat("\n")

# ----------------------------------------
# 2. MERGE (Joining Columns)
# ----------------------------------------
# Scenario: Split job attributes and AI risk
metrics into separate data frames
# and then merge them back using a common
key, 'Job_Title'.

# Dataset A: Job Information (Columns on one
side)
data_job_info <- data_unique_jobs %>%
```

```
  select(Job_Title, Average_Salary,
Years_Experience, Education_Level)

# Dataset B: AI Risk Information (Columns on the
other side)
data_ai_risk <- data_unique_jobs %>%
  select(Job_Title, AI_Exposure_Index,
Automation_Probability_2030, Risk_Category)

print("--- Data A: Job Info (Columns for Merge)
---")
print(data_job_info)
cat("\n")

print("--- Data B: AI Risk (Columns for Merge)
---")
print(data_ai_risk)
cat("\n")

# Perform the MERGE (Inner Join) on "Job_Title"
```

```
merged_data <- merge(data_job_info,
data_ai_risk, by = "Job_Title")

print("--- Merged Data (Columns Added from
Data A and Data B) ---")
print(merged_data)
cat("\n")



# ------------------------------------------
# 3. APPEND (Stacking Rows)
# ------------------------------------------
# Scenario: Take a small subset of the list and
add new, synthetic job entries to it.

# Base List: First 3 jobs from the original list
(selecting only a few columns)
data_base_list <- data_main %>%
  select(Job_Title, Average_Salary,
Risk_Category) %>%
  slice(1:3)
```

```
# New Jobs: Synthetic data with the *same
column structure*
data_new_jobs <- data.frame(
  Job_Title = c("AI Ethicist", "Drone Pilot"),
  Average_Salary = c(150000, 75000),
  Risk_Category = c("Low", "Medium")
)

print("--- Data Base List (First 3 entries) ---")
print(data_base_list)
cat("\n")

print("--- Data New Jobs (For Appending) ---")
print(data_new_jobs)
cat("\n")

# Perform the APPEND (Stacking Rows) using
bind_rows
# Note: bind_rows automatically matches
column names.
```

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```
final_list <- bind_rows(data_base_list,
data_new_jobs)

print("--- Appended Data (New Rows Added) ---")
print(final_list)
```


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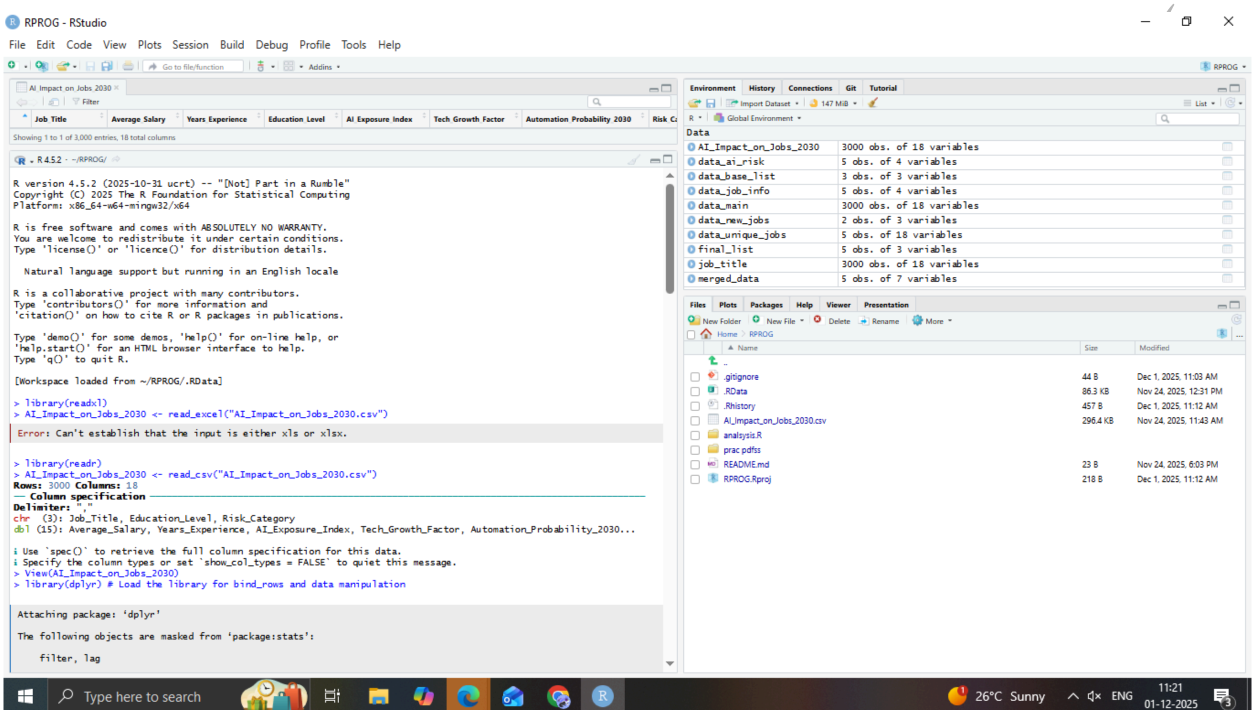

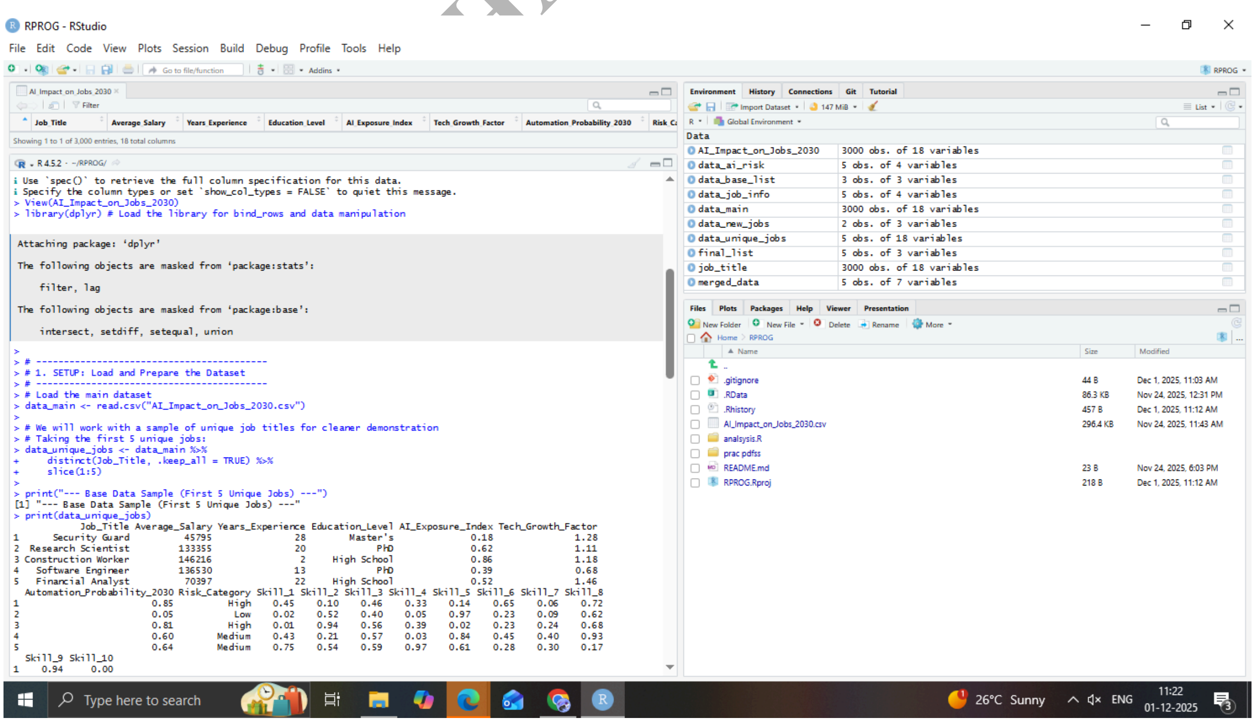

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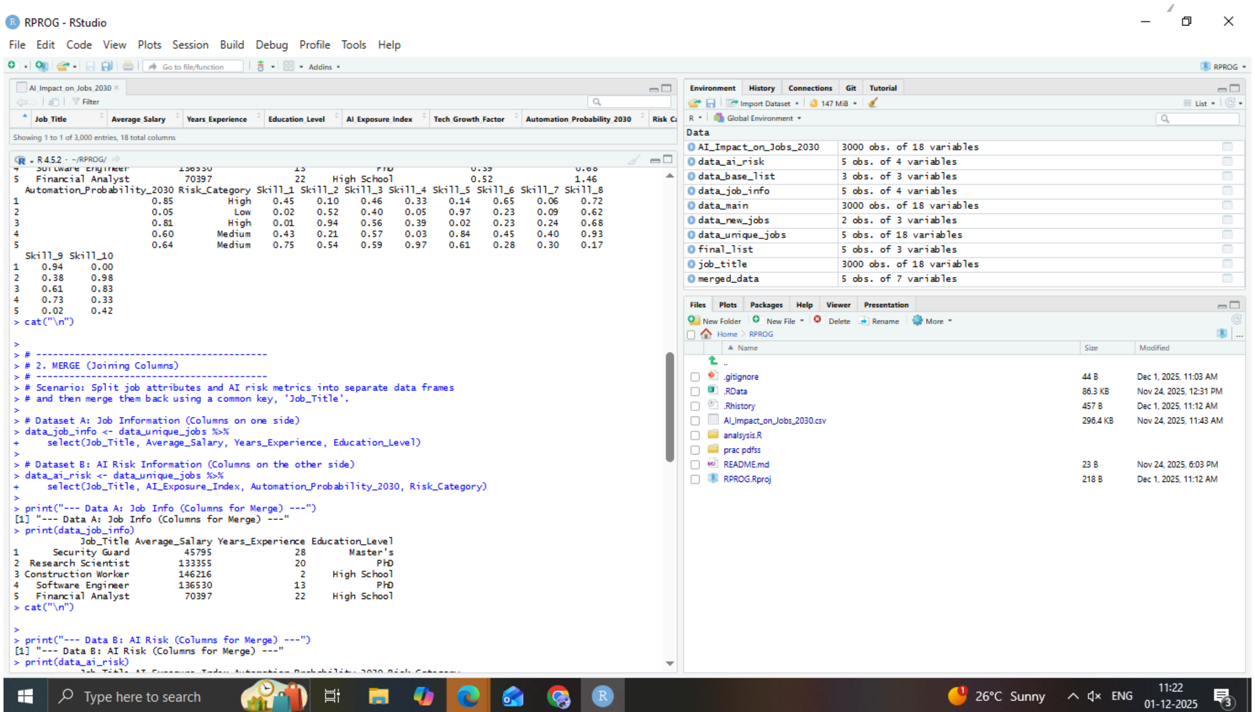

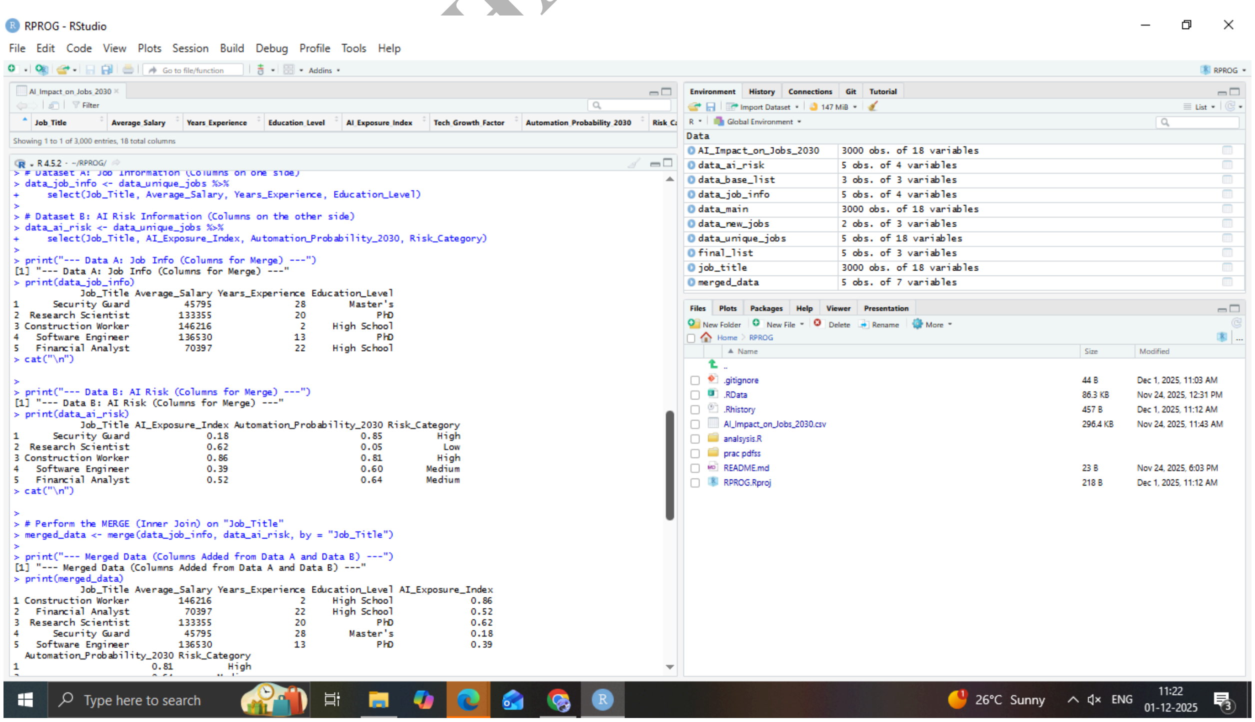

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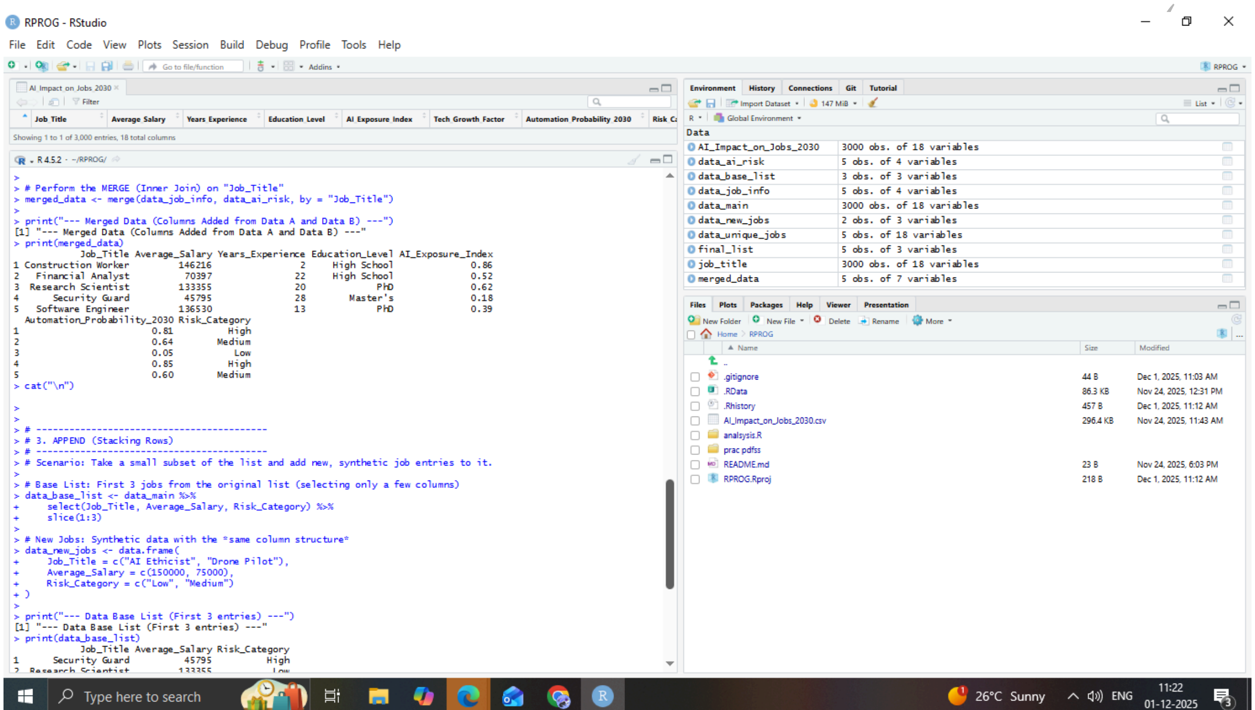

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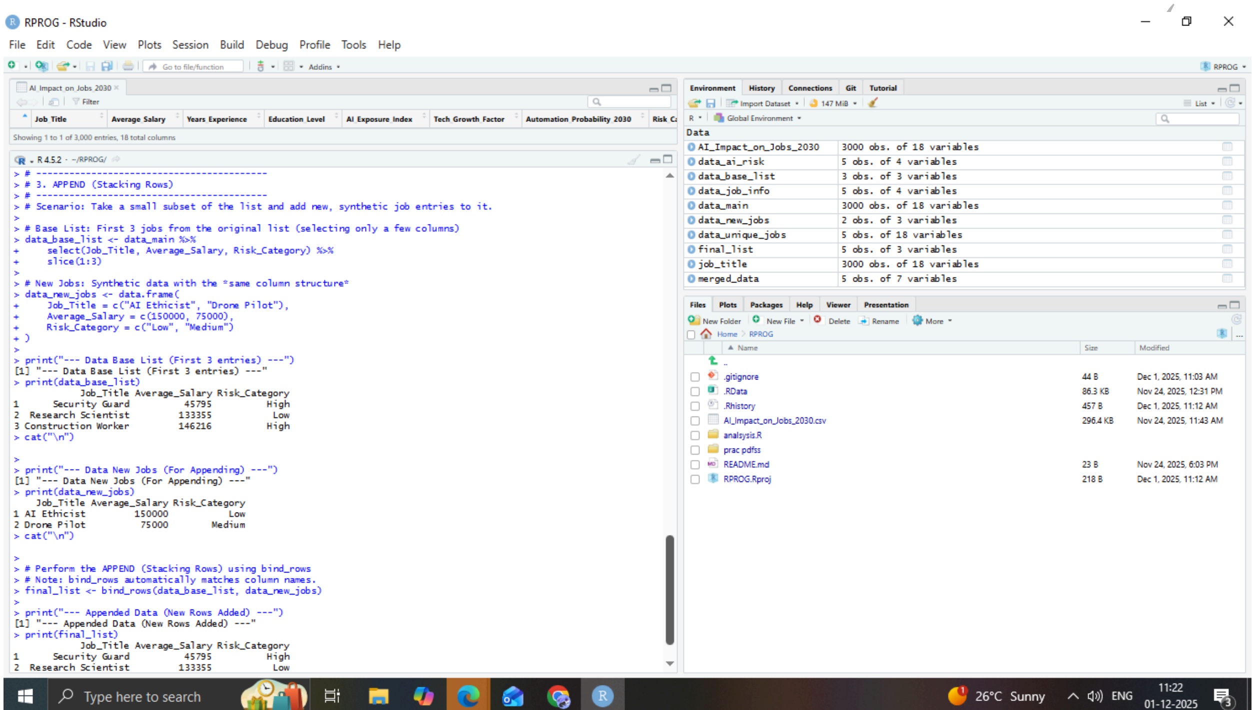

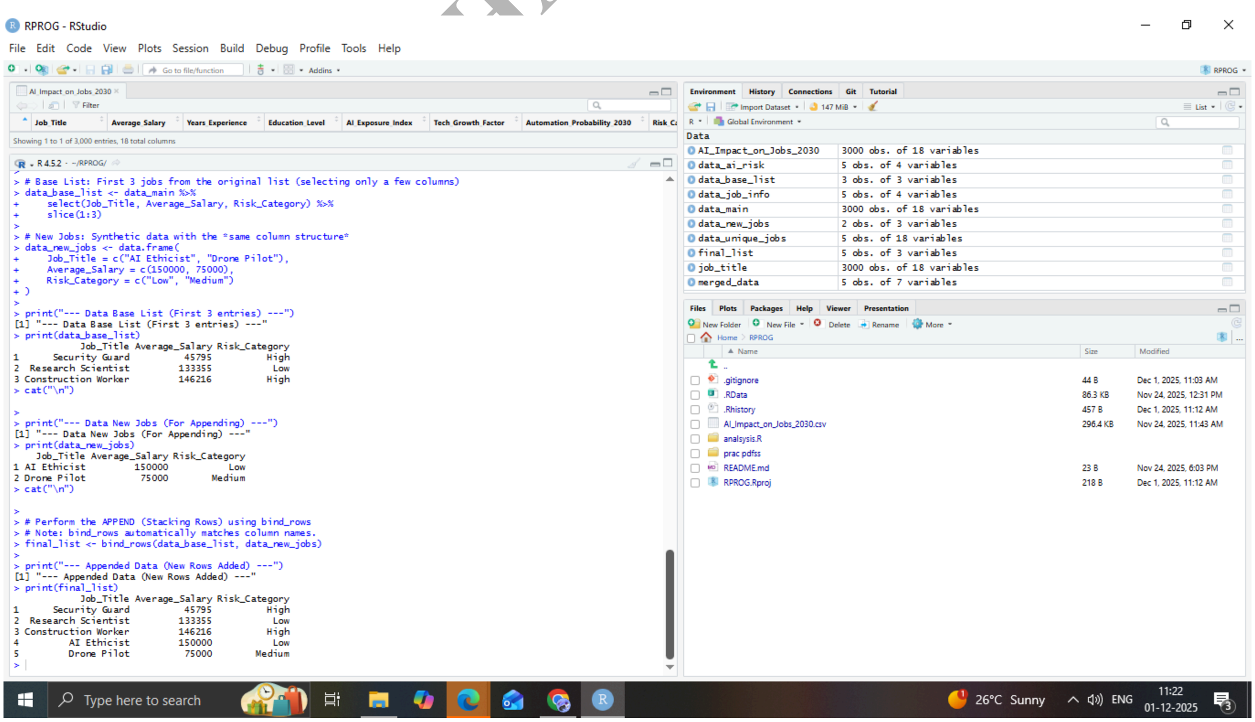

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